Lesson Planning

Complete Lesson Plan Book

Class: Class 8

Subject: Mathematics

Medium: English

Academic Year 2026-27

This book is prepared as a teaching guide for teachers.

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Unit 1: Rational Numbers: Operations

Learning Objectives

- Recall the definition and standard form of rational numbers.
- Add and subtract rational numbers with unlike denominators.
- Multiply and divide rational numbers.
- Solve multi-step problems using the order of operations.

Key Concepts

- A rational number is p/q with q ≠ 0.
- LCM gives the common denominator for addition and subtraction.
- Multiply numerators and denominators directly.
- Dividing means multiplying by the reciprocal.
- BODMAS decides the order when operations are mixed.

Teaching Methodology (Step by Step)

1. Quick revision of standard form with a board drill.
2. Demonstrate addition of 3/4 + 5/6 using LCM 12.
3. Demonstrate division: 7/8 ÷ 14/15.
4. Solve a BODMAS problem step by step.
5. Students solve a graded worksheet: easy to challenging.

Classroom Activities

- Fraction dominoes: match questions to answers.
- BODMAS relay race on the board.

Assessment Questions

1. Compute 5/8 − 7/12 and −9/10 ÷ 3/5.
2. Simplify: 2/3 + 1/2 × 4/5.
3. Write three rational numbers between 1/2 and 3/4.

Homework

- Solve twelve mixed-operation problems on rational numbers.

Time Allocation

Revision 5 min, teaching 20 min, worksheet 10 min, review 5 min.

Unit 2: Exponents and Powers

Learning Objectives

- Define exponents and identify base and exponent.
- Apply the laws of exponents in multiplication and division.
- Simplify expressions with zero and negative exponents.
- Express large numbers in standard (scientific) form.

Key Concepts

- In a^n, a is the base and n is the exponent.
- $a^m \times a^n = a^{m+n}$; $a^m \div a^n = a^{m-n}$.
- Any non-zero number to the power 0 equals 1.
- $a^{-n} = 1/a^n$.
- Standard form writes big numbers as $a \times 10^n$.

Teaching Methodology (Step by Step)

1. Expand 2^5 on the board to show repeated multiplication.
2. Derive the multiplication law with $2^3 \times 2^4$.
3. Derive the division law and the zero-exponent rule.
4. Show negative exponents with $10^{-2} = 1/100$.
5. Convert 4,500,000 to standard form together.

Classroom Activities

- Exponent card game: match laws to examples.
- Write the distances of planets in standard form.

Assessment Questions

1. Simplify: $x^7 \times x^3 \div x^4$.
2. Evaluate 5^{-2} and $(3/4)^0$.
3. Write 0.00056 in standard form.

Homework

- Simplify ten expressions using the laws of exponents.

Time Allocation

Warm-up 5 min, teaching 20 min, practice 10 min, summary 5 min.

Unit 3: Algebraic Identities and Factorisation

Learning Objectives

- State the standard algebraic identities.
- Expand expressions using $(a+b)^2$, $(a-b)^2$ and $(a+b)(a-b)$.
- Factorise expressions by taking out common factors.
- Factorise using identities and grouping.

Key Concepts

- $(a+b)^2 = a^2 + 2ab + b^2$.
- $(a-b)^2 = a^2 - 2ab + b^2$.
- $(a+b)(a-b) = a^2 - b^2$.
- Factorisation is the reverse of expansion.
- Always look for a common factor first.

Teaching Methodology (Step by Step)

1. Multiply $(x+3)(x+3)$ manually to discover the identity.
2. Present the three identities with geometric meaning.
3. Expand five expressions using identities.
4. Factorise $6x^2 + 9x$ by taking out $3x$.
5. Factorise $x^2 - 16$ and $x^2 + 7x + 12$.

Classroom Activities

- Identity bingo: match expanded and factorised forms.
- Factorisation puzzles in pairs.

Assessment Questions

1. Expand $(2x + 5)^2$ and $(3a - 4b)(3a + 4b)$.
2. Factorise: $12xy + 18x$ and $x^2 - 49$.
3. Factorise $x^2 + 9x + 20$.

Homework

- Expand five and factorise five expressions.

Time Allocation

Discovery 5 min, teaching 20 min, practice 10 min, review 5 min.

Unit 4: Linear Equations in One Variable

Learning Objectives

- Solve equations with variables on both sides.
- Solve equations involving brackets and fractions.
- Form equations from word problems.
- Check solutions by substitution.

Key Concepts

- Bring variable terms to one side and numbers to the other.
- Remove brackets first using the distributive law.
- Multiply by the LCM to clear fractions.
- A solution must satisfy the original equation.
- Word problems need a clearly defined variable.

Teaching Methodology (Step by Step)

1. Revise two-step equations with a quick drill.
2. Solve $4x + 7 = 2x + 15$ showing transposition.
3. Solve $3(x - 2) = 2x + 5$.
4. Solve $(x/3) + (x/4) = 14$ using LCM.
5. Translate and solve an age word problem.

Classroom Activities

- Equation stations: rotate through graded problems.
- Make-your-own word problem competition.

Assessment Questions

1. Solve: $5x - 9 = 3x + 11$.
2. Solve: $2(x + 4) - 3x = 10$.
3. The sum of two consecutive numbers is 47. Find them.

Homework

- Solve ten equations including two word problems.

Time Allocation

Drill 5 min, teaching 20 min, stations 10 min, wrap-up 5 min.

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Unit 5: Quadrilaterals and Circles

Learning Objectives

• Classify quadrilaterals by their properties.

• State the angle-sum property of quadrilaterals.

• Define the parts of a circle.

• Solve problems on perimeter and area of composite shapes.

Key Concepts

• Quadrilaterals: square, rectangle, rhombus, parallelogram, trapezium, kite.

• The angles of a quadrilateral add to 360°.

• Parts of a circle: centre, radius, diameter, chord, arc, sector.

• A parallelogram has opposite sides equal and parallel.

• Composite shapes are solved by splitting into known shapes.

Teaching Methodology (Step by Step)

1. Show quadrilateral cut-outs and sort them by properties.
2. Verify the 360° angle sum by tearing corners.
3. Introduce circle parts with a labelled diagram.
4. Solve the area of an L-shaped figure by splitting.
5. Students classify six given quadrilaterals.

Classroom Activities

• Property table completion for six quadrilaterals.

• Draw and label all parts of a circle.

Assessment Questions

1. Three angles of a quadrilateral are 90°, 110°, 80°. Find the fourth.
2. Name the parts of a circle in the diagram.
3. Distinguish between a rhombus and a square.

Homework

• Draw each quadrilateral and write its properties.

Time Allocation

Sorting 5 min, teaching 20 min, problems 10 min, quiz 5 min.

Unit 6: Surface Area and Volume

Learning Objectives

- Define surface area and volume.
- Calculate the surface area and volume of cubes and cuboids.
- Find the volume of cylinders.
- Solve real-life problems on capacity and packing.

Key Concepts

- Surface area is measured in square units; volume in cubic units.
- Cuboid: SA = $2(lb + bh + hl)$; V = $l \times b \times h$.
- Cube: SA = $6a^2$; V = a^3.
- Cylinder: V = $\pi r^2 h$.
- Volume tells how much a container can hold.

Teaching Methodology (Step by Step)

1. Unfold a box to show its six faces (nets).
2. Derive the cuboid surface-area formula from the net.
3. Derive volume as layers: base area × height.
4. Introduce the cylinder volume with a tin can.
5. Solve a water-tank capacity problem.

Classroom Activities

- Build nets of cubes and cuboids from paper.
- Measure a tin and calculate its volume.

Assessment Questions

1. Find the surface area of a cube of side 6 cm.
2. A cuboid is 10 cm × 6 cm × 4 cm. Find its volume.
3. A cylindrical tank has radius 7 m and height 10 m. Find its volume.

Homework

- Solve five problems on surface area and volume.

Time Allocation

Nets 10 min, teaching 20 min, problems 10 min.

Unit 7: Statistics: Mean, Median, Mode and Graphs

Learning Objectives

• Construct frequency tables for grouped data.

• Draw bar graphs, histograms and pie charts.

• Calculate mean, median and mode.

• Interpret data from graphs.

Key Concepts

• Grouped data uses class intervals like 0–10, 10–20.

• A histogram is like a bar graph with touching bars.

• Mean = sum of values ÷ number of values.

• Median is the middle value of ordered data.

• Mode is the most frequent value.

Teaching Methodology (Step by Step)

1. Collect class data on monthly pocket money.
2. Build a grouped frequency table.
3. Draw a histogram step by step.
4. Calculate mean, median and mode of the data.
5. Interpret: what does the average tell us?

Classroom Activities

• Survey and graph: favourite subjects of the class.

• Interpret a newspaper graph in groups.

Assessment Questions

1. Find the mean, median and mode of 12, 15, 12, 18, 20, 12.
2. Draw a histogram for the given frequency table.
3. What is the difference between a bar graph and a histogram?

Homework

• Collect family ages and find mean, median and mode.

Time Allocation

Data 5 min, teaching 20 min, drawing 10 min, interpretation 5 min.

Unit 8: Introduction to Probability

Learning Objectives

• Define probability and its range from 0 to 1.

• List possible outcomes of simple experiments.

• Calculate the probability of simple events.

• Distinguish between certain, likely, unlikely and impossible events.

Key Concepts

• Probability = favourable outcomes ÷ total outcomes.

• Probability ranges from 0 (impossible) to 1 (certain).

• A coin has 2 outcomes; a die has 6.

• Equally likely outcomes have equal chances.

• Experimental probability comes from actual trials.

Teaching Methodology (Step by Step)

1. Toss a coin 20 times and record heads and tails.
2. Define probability with the coin example.
3. List outcomes of rolling a die.
4. Calculate P(even number) and P(7) on a die.
5. Classify events as certain, likely, unlikely or impossible.

Classroom Activities

• Coin-toss experiment and class tally.

• Probability spinners: predict and test.

Assessment Questions

1. Find P(getting a 5) when a die is rolled.
2. What is the probability of getting a head with a coin?
3. Give one example each of a certain and an impossible event.

Homework

• Toss a coin 50 times at home and compare with 1/2.

Time Allocation

Experiment 10 min, teaching 20 min, practice 5 min, review 5 min.